

# NETWORK RECONNAISSANCE & PORT SCANNING AUDIT

RabTech Academy — Cybersecurity & Ethical Hacking Internship

|                   |                                              |
|-------------------|----------------------------------------------|
| Task              | Task 03                                      |
| Assessment        | Network Reconnaissance & Port Scanning Audit |
| Authorized Target | LAB-02 — Metasploitable 2                    |
| Target IP         | 192.168.56.102                               |
| Host Network      | VirtualBox Host-Only Network                 |
| Assessment Date   | 06 October 2026                              |
| Tools             | Nmap 7.991, Wireshark 4.6.9                  |

**Purpose:** Document the authorized reconnaissance, port/service discovery, OS fingerprinting, UDP scanning, and packet-level observations performed against the isolated LAB-02 training system.

**Assessment boundary:** All active scanning in this report was limited to the Metasploitable training VM at 192.168.56.102 on the VirtualBox Host-Only network. No public or third-party systems were targeted and no exploitation was performed.

# 1. Executive Summary

A controlled network reconnaissance assessment was performed against the authorized LAB-02 Metasploitable 2 virtual machine. Connectivity was first verified from the Windows host, followed by TCP SYN scanning, service/version detection, OS detection, and a top-100 UDP scan using Nmap. Wireshark was then used on the VirtualBox Host-Only interface to capture and inspect the reconnaissance traffic.

The assessment identified 23 open TCP ports in the default Nmap scan and 7 UDP ports reported as open or open|filtered in the top-100 UDP scan. Service detection identified multiple legacy services and versions, including Telnet, FTP, old Apache, Samba, rsh/rlogin, MySQL, PostgreSQL, VNC, UnrealIRCd, AJP and Tomcat. Nmap fingerprinted the host as a Linux 2.6.x-family system.

Wireshark confirmed TCP SYN/SYN-ACK reconnaissance traffic between the Windows host (192.168.56.1) and LAB-02 (192.168.56.102). A payload filter returned zero packets in the captured trace, so no application-layer credentials or clear-text payloads were observed in this particular capture. This does not prove that credentials could never be exposed; it only describes the evidence captured during this test.

**Assessment outcome: Completed — evidence collected for Task 03.**

## 2. Scope, Authorization & Lab Topology

### Authorized scope

| Item              | Recorded value                  |
|-------------------|---------------------------------|
| Target            | LAB-02 / Metasploitable 2       |
| Target IP         | 192.168.56.102                  |
| Host-only adapter | 192.168.56.1 / 255.255.255.0    |
| Network type      | VirtualBox Host-Only Network    |
| Scanner           | Windows host running Nmap 7.991 |
| Packet analyzer   | Wireshark 4.6.9                 |
| Exploitation      | Not performed                   |

### Logical topology

```
Windows Host
192.168.56.1
    |
    | VirtualBox Host-Only Network (isolated)
    |
LAB-02 — Metasploitable 2
192.168.56.102
```

The Host-Only design kept the intentionally vulnerable training system separate from the normal Wi-Fi network. The target responded successfully to ICMP from the Windows host.

### 3. Connectivity Verification

Before active scanning, the target was verified as reachable from the authorized Windows host. Four ICMP echo requests were sent and four replies were received, giving 0% packet loss.

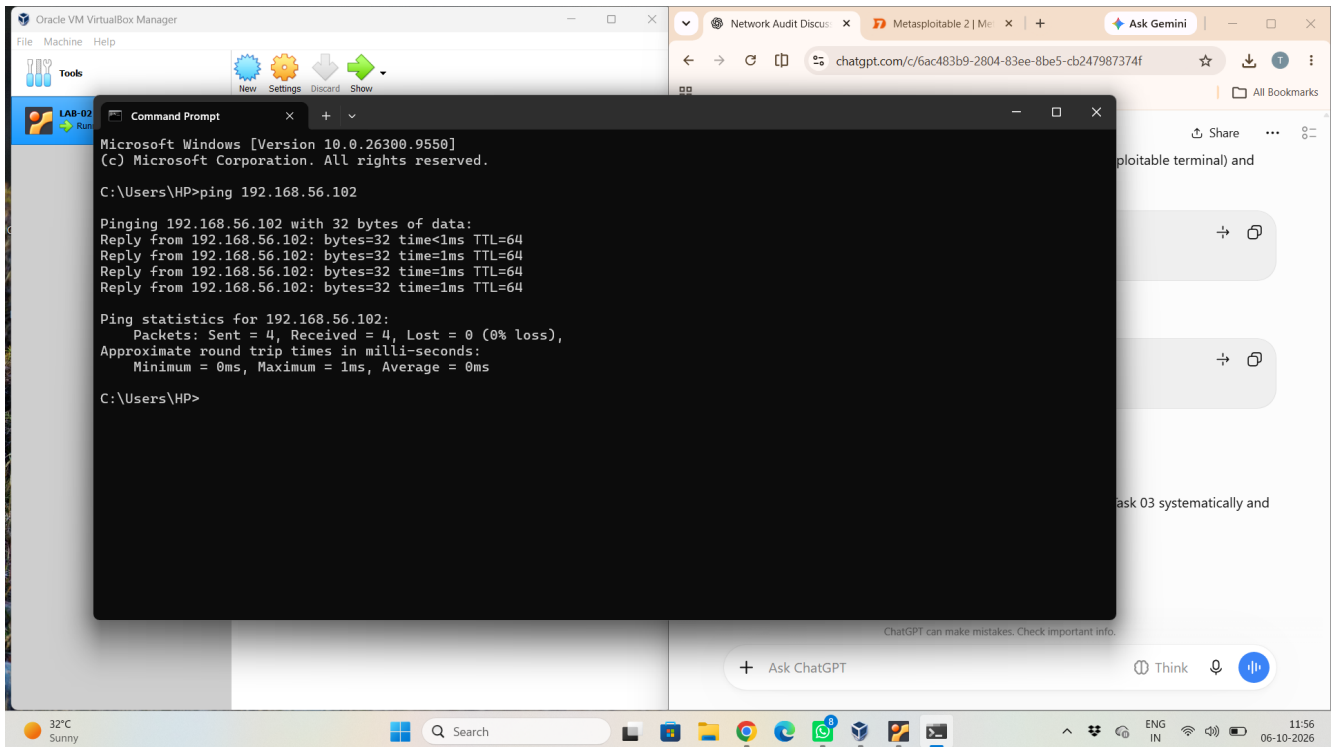


Figure 1. Windows host ping verification of LAB-02 (192.168.56.102): 4/4 replies, 0% loss.

## 4. Nmap Reconnaissance Methodology

The following structured scans were executed against the authorized LAB-02 target. Each output was saved under C:\RabTech\_Task03\nmap\ for evidence preservation.

1) TCP SYN discovery:

```
nmap -sS 192.168.56.102 -oN C:\RabTech_Task03\nmap\tcp_syn_scan.txt
```

2) Service/version detection:

```
nmap -sV 192.168.56.102 -oN C:\RabTech_Task03\nmap\service_version_scan.txt
```

3) OS detection:

```
nmap -O 192.168.56.102 -oN C:\RabTech_Task03\nmap\os_detection.txt
```

4) UDP top-100 scan:

```
nmap -sU --top-ports 100 192.168.56.102 -oN C:\RabTech_Task03\nmap\udp_scan.txt
```

### TCP SYN Scan — Summary

| TCP Port | Detected service |
|----------|------------------|
| 21       | FTP              |
| 22       | SSH              |
| 23       | Telnet           |
| 25       | SMTP             |
| 53       | DNS              |
| 80       | HTTP             |
| 111      | RPCbind          |
| 139      | NetBIOS          |
| 445      | SMB              |
| 512      | exec             |
| 513      | login            |
| 514      | shell            |
| 1099     | RMI registry     |
| 1524     | bindshell        |
| 2049     | NFS              |
| 2121     | FTP              |
| 3306     | MySQL            |
| 5432     | PostgreSQL       |
| 5900     | VNC              |
| 6000     | X11              |
| 6667     | IRC              |
| 8009     | AJP13            |
| 8180     | HTTP/Tomcat      |

The default scan reported 977 closed TCP ports and 23 open TCP ports. The host was up with approximately 0.0057 s latency during the first scan.

## 5. Service & Version Detection Findings

| Port         | Service / Version                              |
|--------------|------------------------------------------------|
| 21/tcp       | vsftpd 2.3.4                                   |
| 22/tcp       | OpenSSH 4.7p1 Debian 8ubuntu1                  |
| 23/tcp       | Linux telnetd                                  |
| 25/tcp       | Postfix smtpd                                  |
| 53/tcp       | ISC BIND 9.4.2                                 |
| 80/tcp       | Apache httpd 2.2.8 (Ubuntu)                    |
| 139, 445/tcp | Samba smbd 3.X–4.X                             |
| 512/tcp      | netkit-rsh rexecd                              |
| 513/tcp      | rlogin / login service                         |
| 514/tcp      | Netkit rshd                                    |
| 1099/tcp     | GNU Classpath grmiregistry                     |
| 1524/tcp     | Metasploitable bind shell / root shell service |
| 2049/tcp     | NFS 2–4                                        |
| 2121/tcp     | ProFTPD 1.3.1                                  |
| 3306/tcp     | MySQL 5.0.51a-3ubuntu5                         |
| 5432/tcp     | PostgreSQL 8.3.0–8.3.7                         |
| 5900/tcp     | VNC protocol 3.3                               |
| 6000/tcp     | X11 (access denied)                            |
| 6667/tcp     | UnrealIRCd                                     |
| 8009/tcp     | Apache JServ (AJP v1.3)                        |
| 8180/tcp     | Apache Tomcat/Coyote JSP engine 1.1            |

**Assessment note:** The identified versions are characteristic of the intentionally vulnerable Metasploitable training image. Version detection establishes attack surface and potential exposure; it does not by itself prove that every listed service is exploitable.

## 6. OS Detection & UDP Findings

### OS fingerprinting

Nmap identified the target as a general-purpose Linux system and estimated the kernel family as Linux 2.6.x, with a reported range of approximately Linux 2.6.9–2.6.33. Network distance was reported as 1 hop.

### UDP top-100 scan

| UDP Port | State         | Service      |
|----------|---------------|--------------|
| 53       | open          | domain / DNS |
| 68       | open filtered | dhcpc        |
| 69       | open filtered | tftp         |
| 111      | open          | rpcbind      |
| 137      | open          | netbios-ns   |
| 138      | open filtered | netbios-dgm  |
| 2049     | open          | nfs          |

The UDP scan completed in 110.65 seconds and reported 93 closed UDP ports among the top 100 probed. Ports marked open|filtered should be treated cautiously because Nmap could not fully distinguish an open service from filtering with the available response behavior.

## 7. Wireshark Packet Analysis

Wireshark was run on the VirtualBox Host-Only Network interface. A controlled Nmap SYN scan against ports 21, 22, 23, 80, 445 and 3306 was performed while the capture was active.

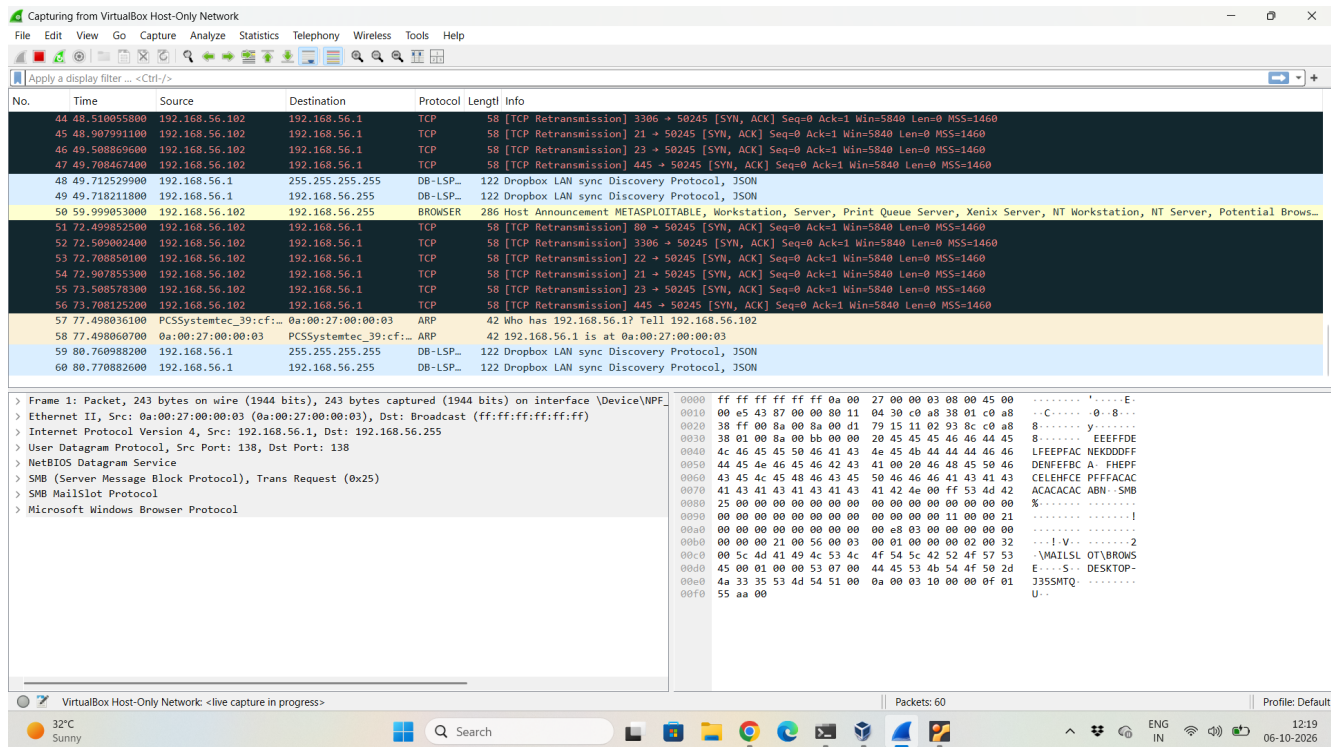


Figure 2. Controlled Nmap scan used to generate clean packet-capture evidence for selected services.

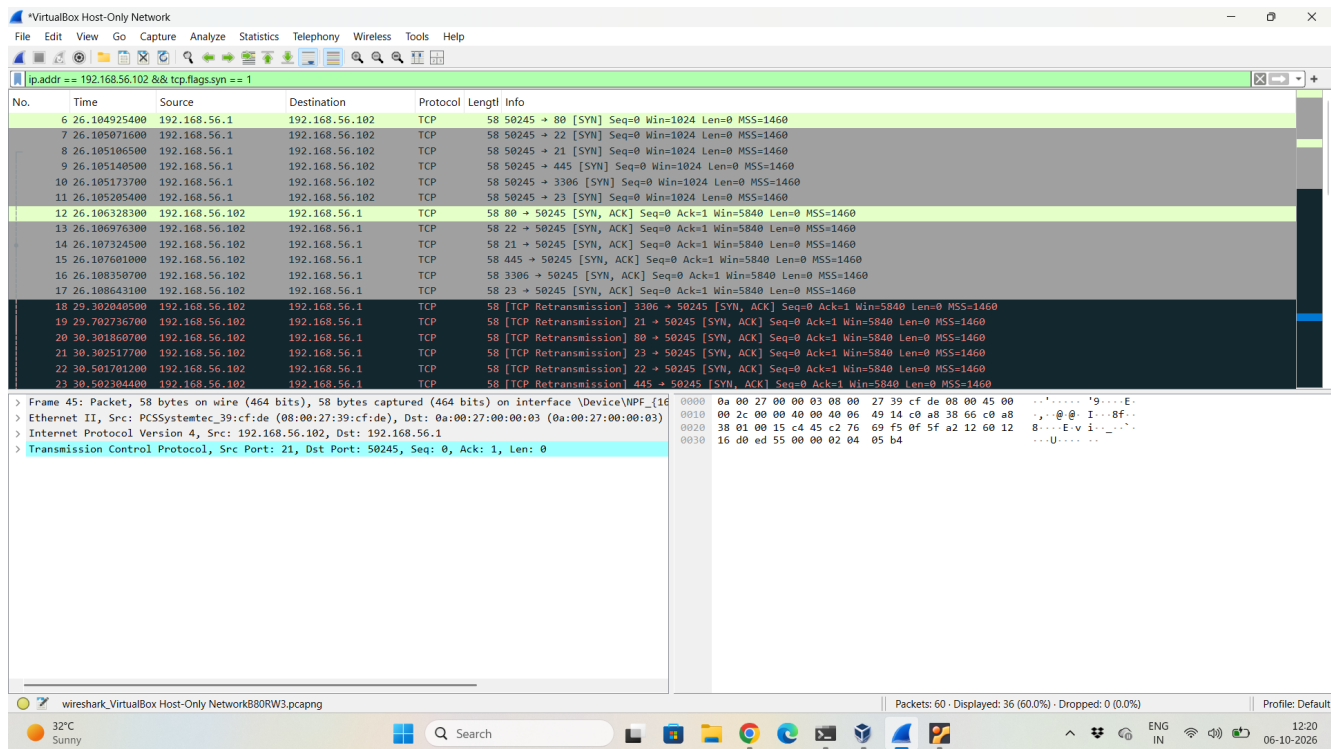


Figure 3. Wireshark display filter showing TCP SYN/SYN-ACK reconnaissance traffic involving LAB-02.

## 8. Clear-Text Payload / Credential Observation

The capture was checked using the display filter:

```
ip.addr == 192.168.56.102 && tcp.len > 0
```

The filter returned zero displayed packets in the saved capture. Therefore, this specific Nmap reconnaissance capture contained no TCP application payload from which credentials could be identified.

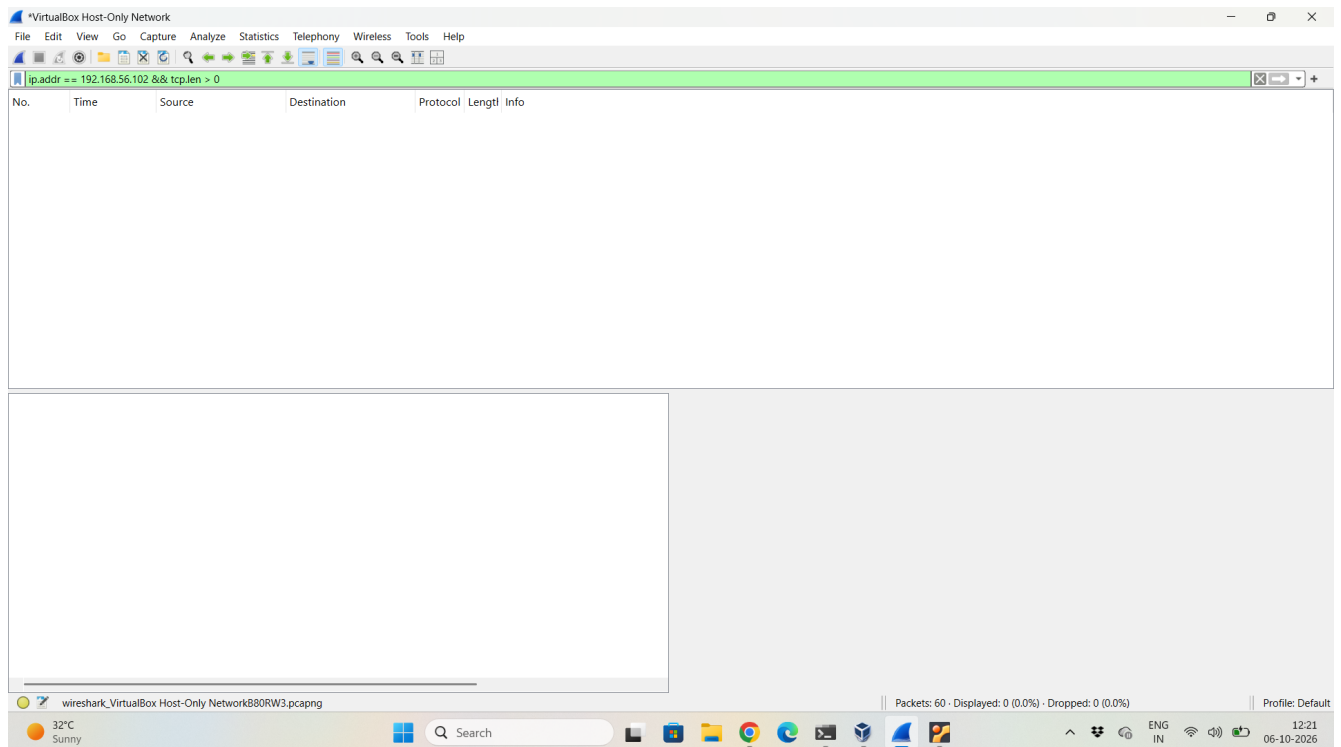


Figure 4. Wireshark payload filter returned 0 displayed packets; no application-layer TCP payload was observed in this capture.

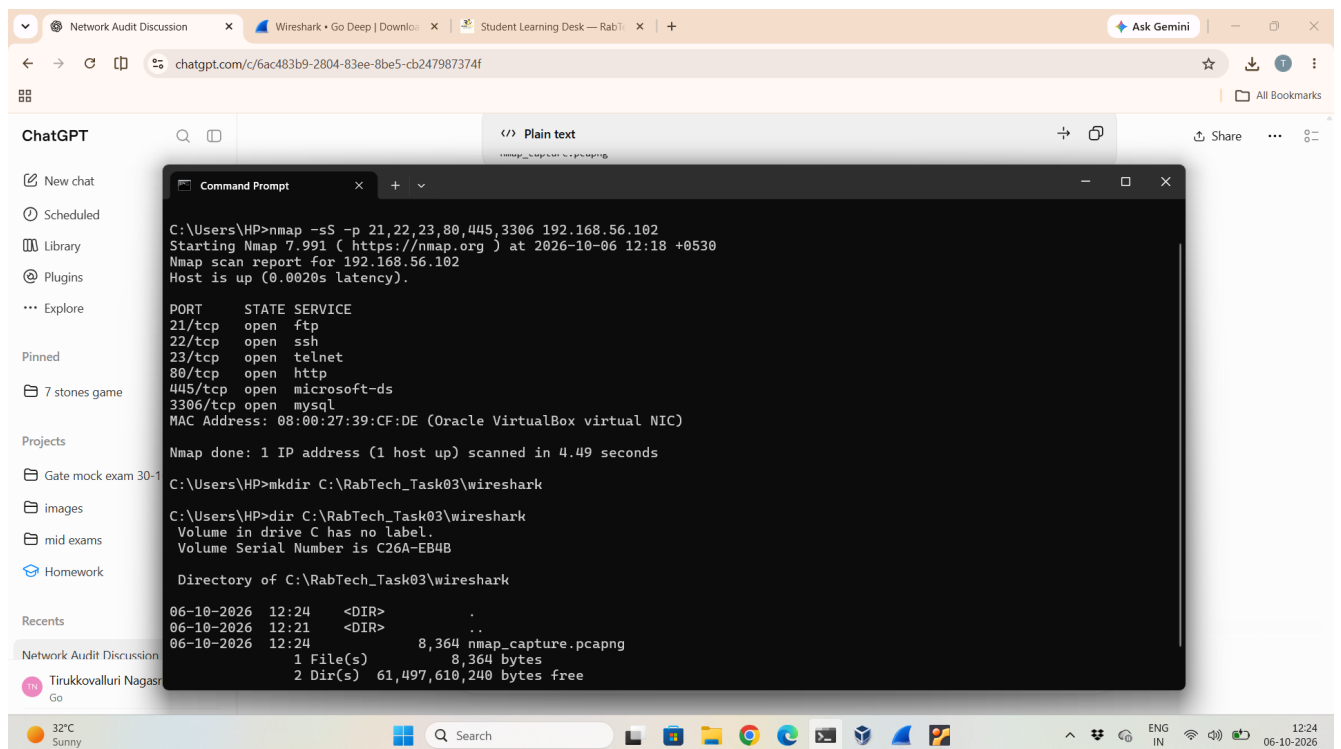


Figure 5. Windows command prompt confirming that nmap\_capture.pcapng was saved under the Task 03 evidence directory.

## 9. Potential Attack Surface & Risk Assessment

The following classifications describe potential security exposure in the intentionally vulnerable training environment. They are not claims that exploitation was attempted or successful.

| Finding                | Potential risk                                                                            | Risk       | Recommended control                                                  |
|------------------------|-------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------|
| Telnet (23/tcp)        | Legacy remote administration can expose credentials and session data if used.             | High       | Disable Telnet; use SSH.                                             |
| FTP services (21/2121) | Legacy/clear-text file transfer increases credential and data exposure.                   | High       | Disable unused FTP; use SFTP/FTPS.                                   |
| rsh/rlogin (512–514)   | Legacy trust-based remote services are unsuitable for modern environments.                | High       | Disable and remove r-services.                                       |
| SMB/Samba (139/445)    | Network file-sharing services expand the remote attack surface.                           | High       | Restrict access, patch Samba, disable unused sharing.                |
| 1524 bind shell        | Service identification indicates a highly dangerous shell exposure.                       | Critical   | Remove/disable; never expose such a service outside an isolated lab. |
| Old web/database stack | Outdated Apache, Tomcat, MySQL and PostgreSQL versions may contain known vulnerabilities. | High       | Patch/upgrade and restrict management/database access.               |
| NFS (2049 TCP/UDP)     | Network filesystem exposure can permit unauthorized access.                               | High       | Restrict exports and network access; use least privilege.            |
| VNC/X11                | Remote graphical services can expose interactive access.                                  | Medium/Low | Disable if unused; restrict and authenticate strongly.               |

**Remediation priorities:** reduce exposed services, remove legacy protocols, patch operating-system and application components, restrict management/database services to trusted networks, and maintain network segmentation for vulnerable systems.

## 10. Evidence Inventory & Reproducibility

The following evidence files were created during the assessment:

| Evidence               | Location                                        | Purpose                              |
|------------------------|-------------------------------------------------|--------------------------------------|
| TCP SYN output         | C:\RabTech_Task03\nmap\tcp_syn_scan.txt         | Open TCP port discovery              |
| Service/version output | C:\RabTech_Task03\nmap\service_version_scan.txt | Service and version identification   |
| OS detection output    | C:\RabTech_Task03\nmap\os_detection.txt         | OS/kernel fingerprinting             |
| UDP output             | C:\RabTech_Task03\nmap\udp_scan.txt             | UDP service discovery                |
| Wireshark capture      | C:\RabTech_Task03\wireshark\nmap_capture.pcapng | Packet-level reconnaissance evidence |

### Reproduction command set

```
nmap -sS 192.168.56.102 -oN C:\RabTech_Task03\nmap\tcp_syn_scan.txt
nmap -sV 192.168.56.102 -oN C:\RabTech_Task03\nmap\service_version_scan.txt
nmap -O 192.168.56.102 -oN C:\RabTech_Task03\nmap\os_detection.txt
nmap -sU --top-ports 100 192.168.56.102 -oN C:\RabTech_Task03\nmap\udp_scan.txt
nmap -sS -p 21,22,23,80,445,3306 192.168.56.102
```

## 11. Conclusion

Task 03 reconnaissance requirements were completed against the authorized, isolated LAB-02 Metasploitable environment. Nmap provided structured TCP, service/version, OS and UDP discovery. Wireshark provided packet-level confirmation of the reconnaissance traffic and a payload check was performed.

The assessment demonstrates a broad attack surface typical of the intentionally vulnerable Metasploitable image, including legacy remote-access protocols, file-sharing services, databases, web services and a shell service. The findings should be interpreted as training-lab evidence, not as an assessment of a production organization.

**Final status: Evidence collection complete.** The report, Nmap outputs, and Wireshark capture can be submitted together according to RabTech Academy's Task 03 submission requirements.

End of Report